Antibacterial effect of ZnS/CdS core-shell nanoparticles

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Abstract*— A core-shell structured ZnS/CdS nanomaterial was synthesized from sol-gel method. Ethylene glycol and thiourea were used as solvent and as sulfur source respectively. Physico-chemical properties of the as synthesized ZnS/CdS core-shell nanoparticles were appraised by PXRD, FTIR, UV-Visible, PL, TEM and EDX analyses. The prepared core-shell material exposes hexagonal wurtsite structure for the CdS and the observed shift in 2θ of the major diffraction peaks towards the lower angle was accredited to the influence of ZnS as the core material. The optical properties were assessed from UV-Visible and Photoluminescence spectral analyses. The emission in the region of 400-550nm elucidates sulfur and metal ion (Cadmium and Zinc) deficiency in the prepared nanoparticles. The FTIR analysis aids to indentify the organic residue in the prepared nanomaterial. The surface morphology of the as synthesized particles was obtained by SEM analysis. The compositional ratio of the sulfur with metal ion was calibrated from EDX analysis. The TEM was utilized to investigate the formation of core-shell structure. The antibacterial test justifies the excellent bacterial inhibition nature of ZnS/CdS core-shell nanoparticle when compared with pure CdS, ZnS and ethanol as solvent.

Keywords*— Sol-gel synthesis, Core-shell materials, Nanoparticles, Antibacterial activity, Metal sulfides.

I. INTRODUCTION

Nowadays semiconducting sulfide nanomaterials were observed much attention due to their valuable functionalities that can differ notably from their bulk counterparts. In accordance with surface to volume ratio, the nanosized metal sulfides consumes large amount of atoms were located at the surface of the nanocrystals when compared with bulk materials [1]. The nanoscaled compounds devote better optical, thermal, mechanical and chemical stability to the as synthesized compounds [2-3]. The core type nano materials composed of two kinds of semiconducting sulfides or selenide counterparts, e.g. CuS/ZnS [4], CdSe/ZnS [5] and CdSe/CdS [6] here the first semiconductor denotes the core material in the as prepared compound while second one was assumed as a shell, the present case the semiconducting sulfide nanomaterials named alike international standards in the literatures. Recent years many techniques were established to fabricate different kinds of core-shell materials with well-refined morphology with reasonable size and good crystallinity. The techniques were named as micro emulsion [7], solvothermal method [8], chemical precipitation technique [9-11], microwave irradiation [12] and sol-gel technique [13]. Among these sol-gel technique offers least expensive in nature, simplicity and ability to capping the particles in nanoscale. The layered arrangements of metal sulfides with one another obtain more impact than composites and alloys and it creates heterostructure to each other successfully [14]. The core-shell structured nanoparticle devotes better inhibition nature against the microbial sources. The metal sulfides have a great attention in drug designing industries for the preparation of new drugs against micro organisms [15-16]. In the present work, core-shell structures comprising ZnS/CdS were synthesized out of sol-gel technique and characterized for their antibacterial applications.

II. EXPERIMENTAL DETAILS

The initial precursor materials were purchased in the form of analytical reagent grade and they were utilized in the synthesis process without any further purification. Required molar ratio of the metal chloride and thiourea were taken as the raw material wherein thiourea was the sulfur source and the ethylene glycol (EG) was used as a solvent. Before going for core-shell structure, the virgin form of CdS and ZnS were synthesized in the present experimental condition by utilizing the starting materials taken in equimolar ratio of metal and sulphur in designated 25ml of ethylene glycol. The core-shell structure was fabricated by adding the solution of 1M of Cadmium Chloride in 25ml of ethylene glycol with the solution of 1M of Zinc chloride and 2M of thiourea in 25ml of ethylene glycol. The excess concentration of sulfur source was taken to balance the metal ions during the formation of ZnS/CdS. The constant stirring of the reaction mixture was carried over for three hours by maintaining the mixture temperature at 85ºC. At the end of the reaction process the ZnS/CdS core-shell product was precipitated as light yellow colored particles. The precipitated product was washed several times with high purity ethanol and filtered (using Whatman filter paper) for the purpose of removing unbounded organic residues. Then, the resultant product was dried in a hot plate to remove the solvent and the dried material was further micro sized with the aid of a agate mortar followed by sintering at 250ºC for 3 hours. Finally, a reasonable

quantity of fine powder of ZnS/CdS compound was obtained. The detailed scheme of the synthesis was shown in Fig 1.

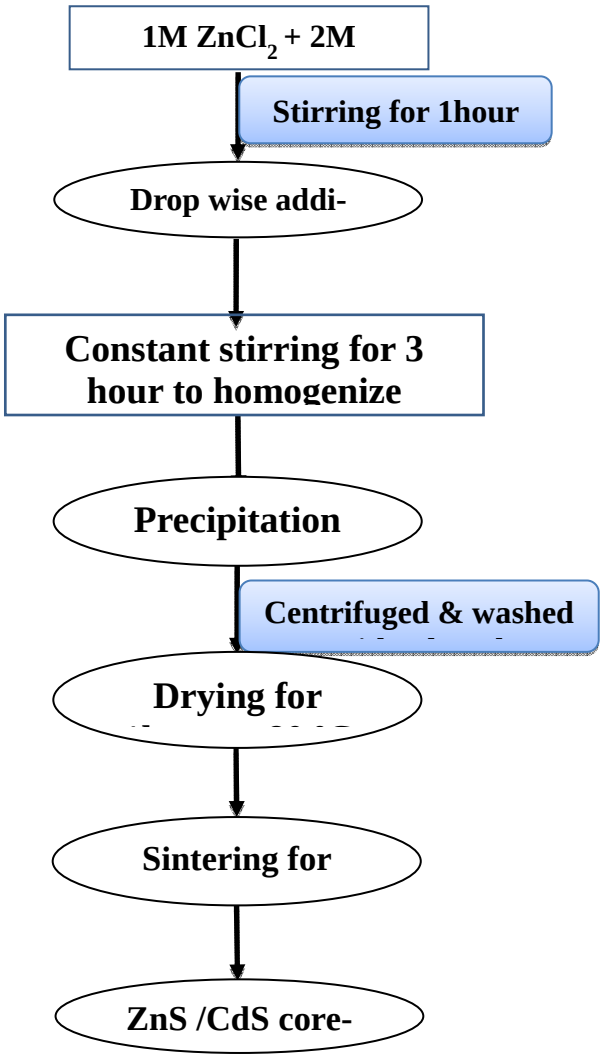


Fig.1. Synthesis scheme of ZnS/CdS core-shell nanoparticles (NPs) by sol-gel technique.

III. Results and Discussion

3.1 XRD Analysis

The crystalline structure of the synthesized nanocrystalline virgin and core-shell samples were estimated from powder X-ray diffraction measurement (PXRD). The prepared nanocrystalline samples exposed different crystal systems. The recorded PXRD patterns of CdS, ZnS and ZnS/CdS core-shell nanomaterials were shown in Fig 2a. The ZnS was synthesized in zinc blende cubic structure (JCPDS Card Number: 65-1691) and the CdS has been synthesized in hexagonal wurtsite (JCPDS Card Number: 80-0006) crystal system [2]. These two diffraction patterns were in good agreement with the standard values. The diffraction data of ZnS/CdS core-shell nanomaterial exposes diffraction peaks alike pure CdS; which is a good evidence for the identification of CdS as a shell material in the prepared core-shell compound [17]. But the major diffraction peaks were shifted towards lower 2theta due to the influence of ZnS nanocrystalline as the core in the compound. The figures 2b and 2c depict the magnified portion of the major diffraction planes ((110), (103), (112), (100), (002) and (101)) of virgin and core CdS particles and their relative shift towards lower angle side.

In Fig.2a the broad diffraction peaks correspond to ZnS indicates that the sizes of the crystallites were comparatively smaller than the other two studied materials. The average crystallite sizes of the prepared CdS, ZnS, ZnS/CdS core-shell nanoparticles (NPs) were estimated from Scherrer's relation (eqn.1) and tabulated (Table 1).

$$D = \frac{0.9\,\lambda}{\beta(cos\theta)} \ldots\ldots\ldots (1)$$

where, D is the crystallite size and λ is the wavelength of monochromatic X-ray radiation (1.54Å). β is full width at half maximum of the diffraction peak and θ is the angle of diffraction of the incident X-ray radiation. The dislocation density (δ) and strain (ε) values of the prepared nanocrystallites were also calculated using the following formulas and were listed in table1.

$$\delta = {}^{1}/_{D^2} \ldots\ldots\ldots (2)$$

$$\varepsilon = \frac{\beta(cos\theta)}{4} \ldots\ldots\ldots (3)$$

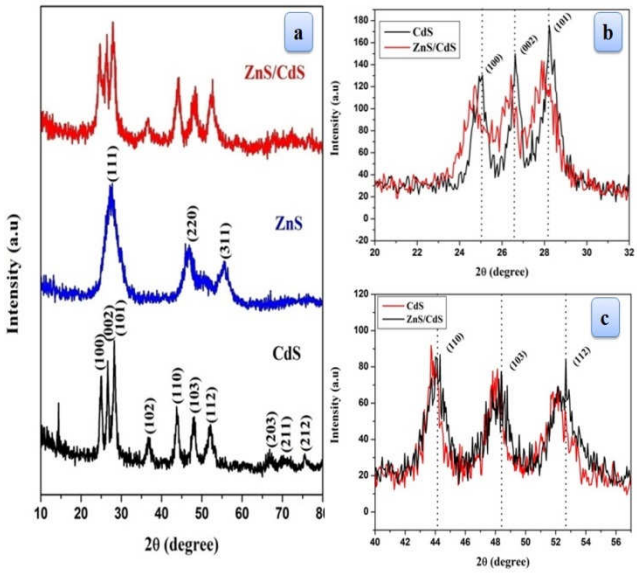


Fig. 2(a) XRD patterns of CdS, ZnS and ZnS/CdS; **(b)** and **(c)** magnified images of the major planes of CdS and ZnS/CdS samples

In the present work, the achieved nanosize of the materials supports the usage of ethylene glycol (EG) as the solvent and also it acted as a good alternative to the conventional capping agents used in the literature [22, 23] to control the size of the synthesized core-shell particles.

Table1

Lattice parameter values of CdS, ZnS and ZnS/CdS nanoparticles

S.No	Sample Name	Crystallite Size (D) (nm)	Dislocation Density(δ) x10^{-4} (Line/m^2)	Strain x 10^{-3} (ε)
1	CdS	32	5.06	3.7
2	ZnS	5	0.04	6.9
3	ZnS/CdS	14	0.53	2.5

3.2. FTIR Measurement

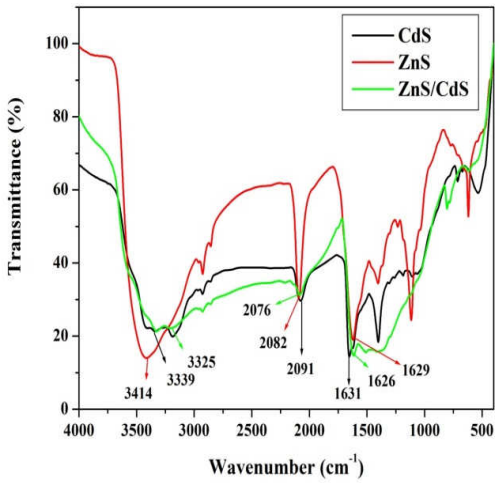


Fig.3. FTIR transmittance spectra for pure CdS, ZnS and ZnS/CdS core shell material.

The consequent molecular vibrations of the as prepared nanoparticles of CdS, ZnS and ZnS/CdS core-shell were assessed from Fourier transform infrared spectral measurement. The FTIR analysis was carried out on the samples pelletized with KBr medium in the wavelength range of 400-4000cm^{-1} and shown in Fig.3. From the transmittance spectrum, the absorption peaks at 3339, 3414 and 3325cm^{-1} are due to the OH stretching molecular vibration of the adsorbed moisture. The molecular vibrational bands between 1627 and 1636 cm^{-1} are due to the stretching vibration of atmospheric CO_2. This kind of adsorption is quite common for nanosized particles due to the high surface area occupancy [18]. In the present case, CO_2 molecular vibrations were appeared at 1629 cm^{-1} (CdS), 1631 cm^{-1} (ZnS) and 1626 cm^{-1} (ZnS/CdS). The characteristic C=C molecular vibrations of organic residues were appeared in the synthesized samples at 2091cm^{-1} (CdS), 2082cm^{-1} (ZnS) and 2076cm^{-1} (ZnS/CdS). Low absorption peaks nearer to 616cm^{-1} has pointed out the formation of CdS and ZnS bonds since this region is commonly assigned to metal-sulphur ion [19]. This weak metal-sulphur band was clearly observable for ZnS than the virgin CdS and ZnS/CdS core-shell. When comparing CdS and ZnS/CdS, the intensity of the metal-sulphur band lies in between virgin CdS and ZnS/CdS and it supports the presence of ZnS in core-shell compound [13, 20].

I.*3.3. UV-Visible spectral Analysis*

The UV-Visible absorption and transmittance spectra of the as prepared CdS, ZnS and ZnS/CdS were shown in figures 3a and 3b. The synthesized nanoparticles exhibit strong absorption edges around 537, 377 and 976nm for CdS, ZnS and ZnS/CdS core-shell respectively. In literature, lesser particle size exposes higher order of absorption edge [20]. This fact is justified in the case of ZnS and ZnS/CdS core-shell. Further, ZnS/CdS core-shell has resulted a diffused absorption like ZnS rather than sharp absorption like CdS and red shifted. This may be attributed to the size as well as potential–well effect. When a shell was formed over the surface of the core, the size of the shell is higher than that of the core material. This allows further delocalization in electronic wave function and make a red shift in band edge transition (size effect) [21]. The optical absorption co-efficient (α) of the prepared CdS and ZnS nanoparticles were calculated using the following relation.

$$\alpha = \frac{2.3036\log(1/T)}{d} \ldots\ldots\ldots (4)$$

Where T is the transmittance value of the as prepared nanoparticles.

As direct band gap, the nanoparticle under measurement has absorption co-efficient (α) obeying the following expression at high photon energies.

$$\alpha = \frac{A(hv - E_g)^{1/2})}{hv} \ldots\ldots\ldots (5)$$

Where E_g is the optical energy gap, A is a constant. The optical band gap E_g was evaluated from the following equation as 2.31eV and 3.29eV for CdS and ZnS respectively.

$$E_g = \frac{1240}{\lambda} \;\; in\ eV \ldots\ldots\ldots (6)$$

Where 'λ' is the cutoff wavelength. The optical band gap was also evaluated by extrapolating the linear portion of plots of $(\alpha h\upsilon)^2$ and (hυ) as shown in Fig 4c [18].

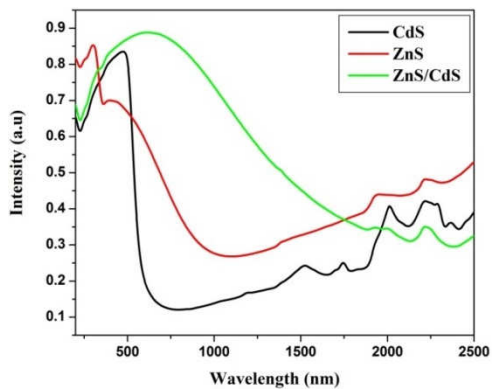


Fig.4a. Absorption spectrum of CdS, ZnS and ZnS/CdS nanoparticles.

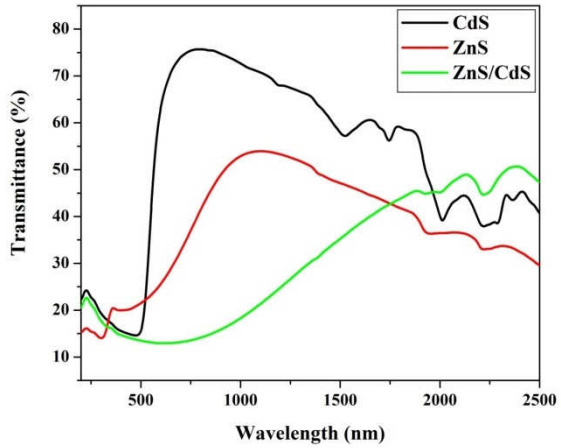


Fig.4b. Transmittance spectrum of CdS, ZnS and ZnS/CdS nanoparticles

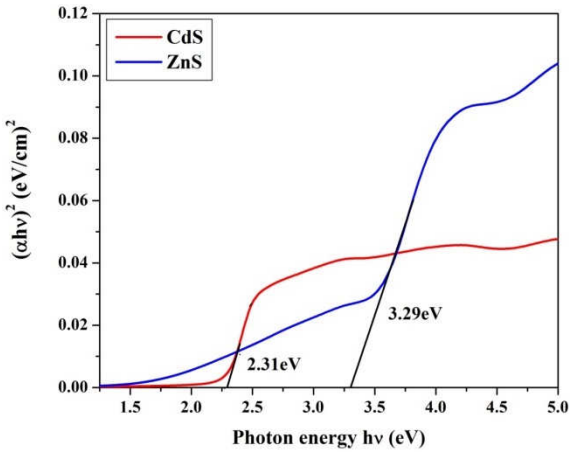


Fig.4c. Tauc's plot image of the as prepared CdS and ZnS nanoparticles

3.4 Photoluminescence Measurement

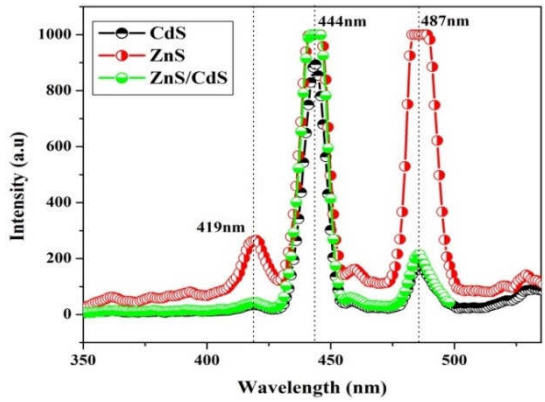
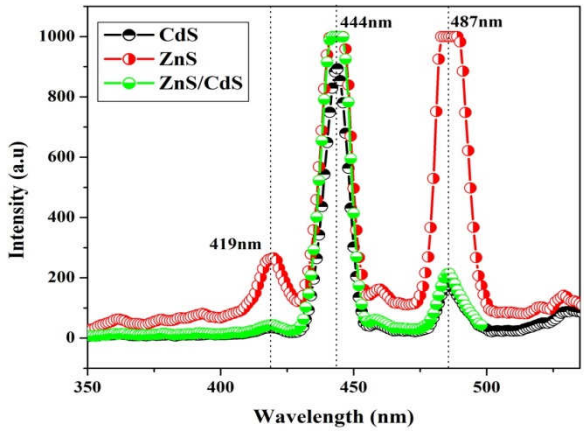


Fig.5. Photoluminescence spectrum of CdS, ZnS and ZnS/CdS nanoparticles

Fig.5 shows the emission spectra of CdS, ZnS and ZnS/CdS core-shell nanoparticles. According to the literature and from the recorded UV-Visible absorption spectral data, the studied samples were excited at 220nm [13]. Three emission peaks at 419, 444 and 487nm were observed for the studies samples, out of which the emission at 444nm was intense for all the three samples and the emission at 487nm was intense for ZnS alone. The strong emission at 444 and 487nm can be attributed to the electron-hole pair recombination in sulfur and metal ion vacancies whereas the emission at 419nm may related to other surface related defects [18]. The emission at 419nm for ZnS may give rise to an increase in the intensity of the emission at 444nm for the ZnS/CdS due to its secondary excitation (primary excitation was at 220nm). But the strong emission at 487nm for the virgin ZnS may not contribute in the ZnS/CdS since the CdS shell layer is almost transparent for this wavelength. This observation on ZnS/CdS evidences the formation of core-shell structure [22, 20].



3.5 *SEM Analysis*

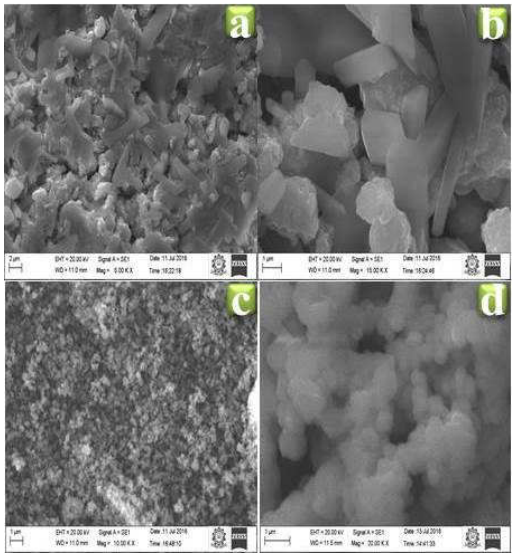


Fig.6. SEM micro graphs of **(a, b)** CdS; **(c)** ZnS; **(d)** ZnS/CdS nanoparticles

The SEM images of CdS, ZnS and ZnS/CdS were shown in Fig.6. The microscopical images in 6a and 6b for CdS at different magnifications recommends the formation of nanorod like structure and agglomerated particles. This may endorse the large number of multinucleations and extensive reaction time, sintering temperature and its duration in the experimental process. Fig.6c shows the distinctive morphological image of virgin ZnS nanoparticle having smaller dimen-

sion when compared with CdS. The smaller dimension (i.e., 5nm from PXRD) ascertains the efficacy of the adopted process to synthesize ZnS nanomaterial using ethylene glycol as the solvent as well as the capping agent and thiourea as the sulphur source. Fig.6d depicts the SEM image of ZnS/CdS core-shell particles which comprises uniform sphere like morphology [23] in the range of nanometer. The evenly appearing particles favour the optimized experimental conditions, the calibrated quantity of initial precursors and the good capping property of the ethylene glycol apart from acted as solvent [17, 24].

3.6 EDAX Measurement

From the EDAX analysis, the presence of elements and its molecular compositions were intended. Fig.7(a, b, c) shows the energy dispersive X-ray spectra for CdS, ZnS and ZnS/CdS core-shell nanoparticles and table 2 lists out the compositional ratios of the synthesized samples. The absence of new elements except oxygen supports the purified form of the synthesized material. The appearance of oxygen raised the alarm on the high sintering temperature during the process which promoted the metal-oxide formation. But the atomic ratio of the oxygen in the sample was very negligible when compared with the product materials [25-27].

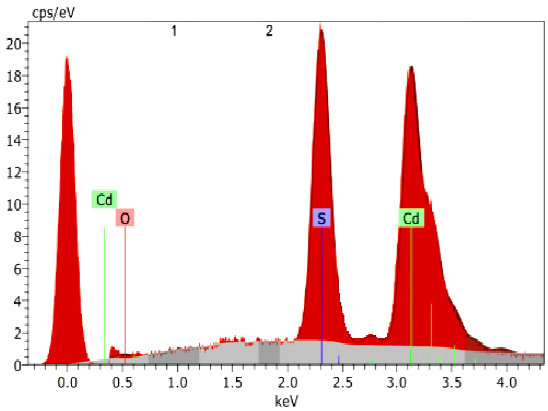


Fig.7a. EDAX spectrum of CdS nanoparticle

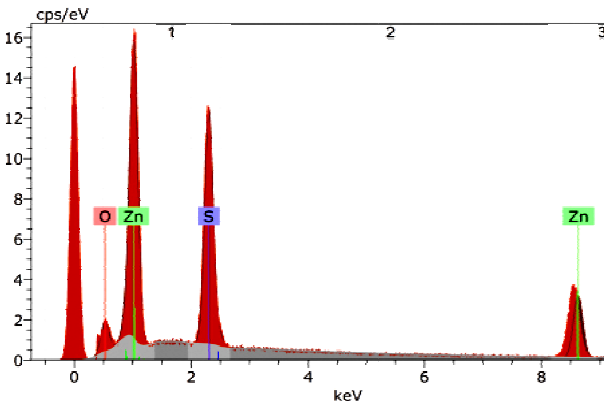


Fig.7b. EDAX spectrum of ZnS nanoparticle

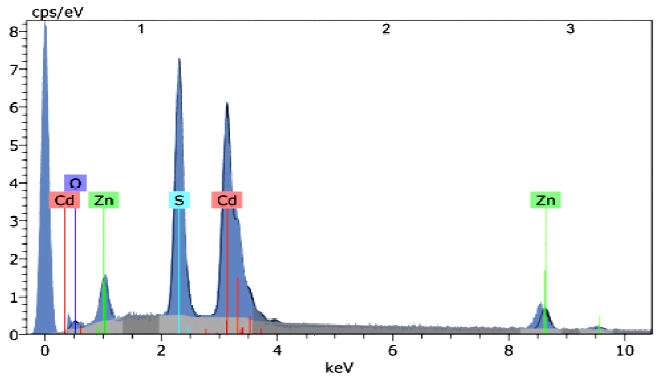


Fig.7c. EDAX spectrum of ZnS/CdS core-shell nanoparticles

Table 2

Composition ratios of as synthesized nanoparticles

S.No	Elements	ZnS/CdS wt(%)	CdS wt(%)	ZnS wt(%)
1	Cadmium	56.88	63.66	-
2	Zinc	15.98	-	57.65
3	Sulfur	24.58	32.66	40.18
4	Oxygen	2.56	3.68	2.17
	Total (%)	100	100	100

3.7 TEM Analysis

TEM images of ZnS/CdS under different magnifications were exhibited in figures 8 (a-c) and 9. The estimated particle size was ~ 10-16nm and it very well matches with the calculated crystallites size (14nm) from PXRD. The noticeable boundary with defined lattice strains establishes the core-shell nature of the ZnS/CdS and the SAED pattern supports the high degree of crystallinity in the grains [26, 29].

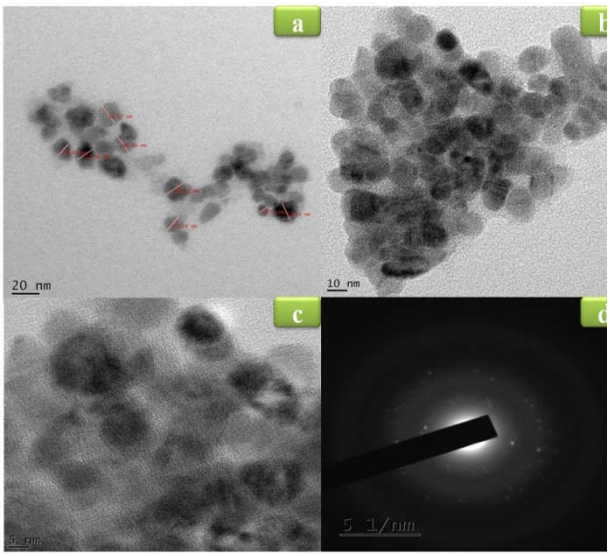


Fig.8 (a,b,c) Different magnified scale ranges of TEM images of ZnS/CdS core-shell material; **(d)** SAED pattern of the ZnS/CdS core-shell material

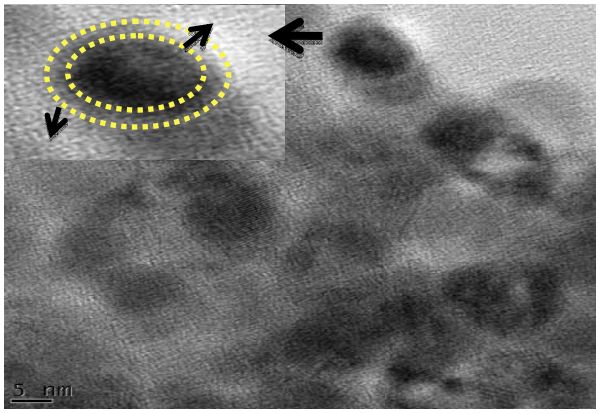


Fig.9. Magnified portion of the single core-shell from TEM image

3.8 Antibacterial Activity

Normally the semiconducting metal sulphides exhibits better inhibition against micro-organisms [15, 16] in that view the given as synthesized metal sulphide compounds were subjected to the antibacterial studies. The antibacterial activity was obtained by the assist of one gram positive (Enterococcus faecalis) and one gram negative (Pseudomonas aeruginosa) bacteria's. Fig.10 shows the bacterial culture inhibition images of ethanol, CdS, ZnS and ZnS/CdS materials. The solvent ethanol has the antibacterial property in higher concentration and the prepared nanocrystalline drugs of CdS, ZnS and ZnS/CdS core-shell materials exhibit a better antibacterial activity, when compared with the dispersion solvent ethanol. The prepared nano structured metal sulfide particles were dispersed thoroughly in ethanol by ultra sonication process. The bacterial culture was developed in the petri dishes; the dispersed metal sulfides in different concentration were pointed into the culture dishes. The pure and core CdS material express the better inhibition nature against the gram (+) as well as gram (-) bacterias. The pure ZnS exhibit good inhibition nature against the gram (+) bacteria but in lower concentrations it does not have any significant antibacterial property against the gram (-) bacteria. This is a good evidence to identify the CdS was occupied as a shell position in the as synthesized core-shell structure. Corresponding concentration dependent inhibition level of the given bacterial culture details were listed in table 3 [30].

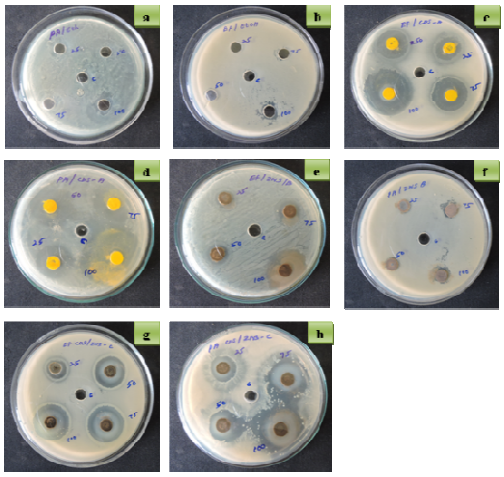


Fig.10. Bacterial inhibition images of ethanol, CdS, ZnS and ZnS/CdS materials

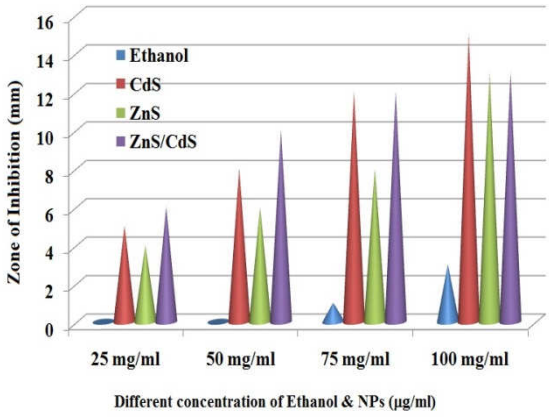


Fig.11a. Bacterial inhibition comparative plot of Gram (+) bacteria

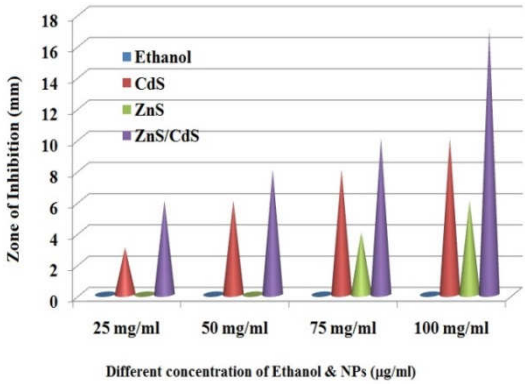


Fig.11b. Bacterial inhibition comparative plot of Gram (-) bacteria

Table 3

Concentration dependent inhibition level of the given bacterial culture details

S.No	Sample code	Microbial Culture Name	Microbial Details Accession Number	Concentration	Zone Of Inhibition Level (mm)
1.	CdS	Gram (+) *Enterococcus faecalis*	Accession number:HQ693279.1 DahE1Dr.B.Vaseeharan lab	25µg/ml	5mm
				50 µg/ml	8mm
				75 µg/ml	12mm
				100 µg/ml	15mm
		Gram (-) *Pseudomonas aeruginosa*	ATCC: 27853	25µg/ml	3mm
				50 µg/ml	6mm
				75 µg/ml	8mm
				100 µg/ml	10mm
2.	ZnS/CdS	Gram (+) *Enterococcus faecalis*	Accession number:HQ693279.1 DahE1Dr.B.Vaseeharan lab	25µg/ml	6mm
				50 µg/ml	10mm
				75 µg/ml	12mm
				100 µg/ml	13mm
		Gram (-) *Pseudomonas aeruginosa*	ATCC: 27853	25µg/ml	6 mm
				50 µg/ml	8 mm
				75 µg/ml	10 mm
				100 µg/ml	17 mm
3.	ZnS	Gram (+) *Enterococcus*	Accession number:HQ693279.1	25µg/ml	4mm
				50µg/ml	6mm

IV. Conclusion

The metal sulfides of virgin CdS, ZnS and ZnS/CdS core-shell were synthesized from sol gel technique. The structural properties were established from powder X-ray diffraction analysis which shows CdS belongs to , ZnS belongs to and Zns/CdS resembles the structure of CdS but with a 2theta shift towards lower angle. Further the more structural information on the characteristics molecular vibrations of the samples were assessed from FTIR analysis. The optical properties acquired from UV-Visible and photoluminescence analyses have evidence the cut-off frequency and 3 emission peaks. The sharp cutoff wavelengths of 537nm for CdS was observed whereas a diffused slope was seen in ZnS and ZnS/CdS. The excitation of the samples at 220nm has resulted optical emissions at 419,444 and 487nm. The synthesized nanoparticles expressed well refined morphology in SEM analysis which articulates the suitability of the precursor materials and the capping ability of the process. The chemical composition was calculated by EDAX analysis. The TEM analysis shows that the synthesized ZnS/CdS core-shell particles was in the range of 10-16nm. The noticeable boundary with defined lattice strains establishes the core-shell nature of the ZnS/CdS and the SAED pattern supports the high degree of crystallinity in the grains. The antibacterial properties of the CdS, ZnS and ZnS/CdS were carried out. The virgin CdS and core-shell ZnS/CdS has the better bacterial inhibition nature than virgin ZnS and ethanol.

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Figure captions

Fig.1. Synthesis scheme of CdS/ZnS nanoparticles by sol-gel technique.

Fig.2(a) XRD pattern of CdS, ZnS and ZnS/CdS; **(b)** and **(c)** magnified images of major planes in pure and core ZnS/CdS samples.

Fig.3. FTIR transmittance spectra for pure CdS, ZnS and CdS/ZnS core shell material.

Fig.4a. Absorption spectrum of CdS, ZnS and ZnS/CdS nanoparticles.

Fig.4b. Transmittance spectrum of CdS, ZnS and ZnS/CdS nanoparticles.

Fig.4c. Tauc's plot image of the as prepared CdS, ZnS and ZnS/CdS nanoparticles.

Fig.5. Photoluminescence spectrum of CdS, ZnS and ZnS/CdS nanoparticles.

Fig.6. SEM micro graphs of **(a, b)** CdS; **(c)** ZnS; **(d)** ZnS/CdS nanoparticles.

Fig.7a. EDAX spectrum of as synthesized CdS nanoparticle.

Fig.7b. EDAX spectrum of as synthesized ZnS nanoparticle.

Fig.7c. EDAX spectrum of Core-shell ZnS/CdS nanoparticles.

Fig.8 (a,b,c) Different magnified scale ranges of TEM images of ZnS/CdS core-shell material; **(d)** SAED pattern of the ZnS/CdS core-shell material.

Fig.9. Magnified portion of the single core-shell from TEM image.

Fig.10. Bacterial inhibition images of ethanol, CdS, ZnS and ZnS/CdS materials.

Fig.11a. Bacterial inhibition comparative plot of Gram (+) bacteria.

Fig.11b. Bacterial inhibition comparative plot of Gram (-) bacteria.

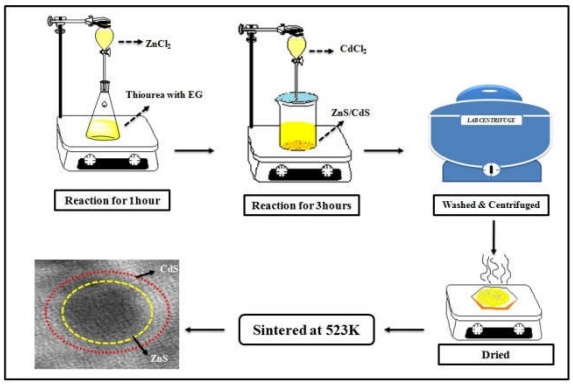
Table captions

Table 1 Lattice parameter values of CdS, ZnS and ZnS/CdS nanoparticles.

Table 2 Composition ratios of as synthesized nanoparticles.

Table 3 Concentration depended inhibition level of the given bacterial culture details.

Research highlights

- The ZnS/CdS core-shell nanoparticles were synthesized in least expensive sol-gel technique.
- The as synthesized ZnS/CdS core-shell compound express CdS compound related diffraction data which was a good evidence to know the CdS is a shell material for the given core-shell compound.
- From microscopical measurement the as prepared ZnS/CdS core-shell compound expose uniformed sphere like morphology.
- According to TEM analysis the given core-shell compound is in the range for few tens of nanometer.
- The as synthesized ZnS/CdS core-shell nanoparticle devotes very good inhibition nature against bacterial micro-organisms.

Graphical abstract